

ORIGINAL ARTICLE

Abdominal Muscle Performance as Measured by the Double Leg-Lowering Test

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ABSTRACT. Krause DA, Youdas JW, Hollman JH, Smith J. Abdominal muscle performance as measured by the double leg-lowering test. *Arch Phys Med Rehabil* 2005;86:1345-8.

Objective: To describe abdominal muscle performance as measured by the double leg-lowering test (DLLT) in healthy subjects.

Design: Repeated measures.

Setting: Academic laboratory.

Participants: A sample of 100 healthy volunteers (50 men, 50 women; age range, 18–29y).

Interventions: Not applicable.

Main Outcome Measures: Repeat measurements of the DLLT to determine reliability and performance standards and to identify variables that predict performance.

Results: The intraclass correlation coefficient ($ICC_{3,1}$) for repeated measures of the DLLT was .98. The mean value $\pm$ standard error of the mean for abdominal performance for men was $15.4^\circ \pm 2.3^\circ$ from a horizontal reference, and for women it was $36.9^\circ \pm 3.4^\circ$. A significant difference was found between men and women on performance of the DLLT ($t_{98} = -5.3$, $P < .001$). A linear regression model found sex ($t = -3.2$, $P = .002$) and age ($t = -4.6$, $P < .001$) to be significant predictors of performance on the DLLT.

Conclusions: The DLLT has excellent intratester reliability. We also found a sex difference in the performance of the DLLT.

Key Words: Evaluation studies; Leg; Muscles; Rehabilitation.

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ABDOMINAL MUSCLE PERFORMANCE is frequently examined in the clinical setting to help guide therapeutic interventions. In addition to describing the traditional sit-up or abdominal curl, several researchers have described the double leg-lowering test (DLLT) to assess and monitor abdominal strength.¹⁻⁴ Whereas the abdominal curl test assesses the ability of the abdominal muscles to flex the trunk against gravity with respect to the pelvis, the DLLT assesses the ability of these muscles to stabilize the pelvis in a posterior-tilted position against an external load imparted by the lower extremities as they are lowered from a vertical starting position.

Proper DLLT performance requires palpation of a bony landmark to ensure that the abdominal muscles are functioning in a stabilizing role to maintain the pelvis in a position of posterior tilt as the legs are lowered. Without monitoring, the pelvis could be in a position of anterior tilt with the lumbar spine extended, resulting in hip flexor muscles performing the leg-lowering movement without a stabilizing contribution from the abdominal muscles.⁵ This aspect has an important clinical application because core stability and lumbar stabilization programs rely on the abdominal muscles functioning effectively in a stabilizing role.⁶ Currently there are limited objective criteria with which to measure core performance.

Kendall et al⁷ contended that men should obtain a DLLT grade of “normal” (0° , or complete lowering without loss of pelvic control) and women a grade of “good” (30° from horizontal before loss of pelvic control). However, this grading system has not been validated. Youdas et al⁸ reported DLLT mean values in a group of asymptomatic adults 40 to 70 years old. On average, men were able to lower their legs to a position of $39.4^\circ \pm 11.3^\circ$ from a horizontal reference and women to a position of $49.6^\circ \pm 11.5^\circ$. Consequently, although clinicians use the DLLT to examine abdominal muscle performance and as an exercise in intervention programs, normative performance values for young adults have not been reported.

The purpose of our study was to describe abdominal muscle performance as measured by the DLLT in healthy men and women between ages 18 and 29 years for use as a component of a clinical assessment of core function. In addition, we investigated the potential influence of 9 factors (sex, age, leg length, height, body mass, physical activity level, regular program of physical activity, regular program of abdominal strengthening, frequency of abdominal strengthening) on abdominal performance as measured by the DLLT.

METHODS

Participants

Fifty men and fifty women between 18 and 29 years old volunteered to participate in the study. We selected subjects in this age range because they often are evaluated in our sports medicine clinic for low back pain (LBP). Also, it is in this age group that abdominal strengthening is commonly incorporated for improving core performance. The mean $\pm$ standard deviation (SD) age of the men was 24.8 ± 2.4 years and of the women was 22.5 ± 2.5 years. The mean heights of the men and women were 181.6 ± 6.6 cm and 166.4 ± 6.4 cm, respectively, and their mean weights were 86.1 ± 16.2 kg and 62.0 ± 7.8 kg. Exclusionary criteria included current known pregnancy, past spinal operations, current LBP, or any self-reported hip abnormality that limited hip range of motion. Testing was not performed within 1 hour after a meal. Volunteers were recruited through personal contact by the authors and through advertisements. Informed consent to participate was obtained from all volunteers. The study was approved by the institutional review board of the authors' institution.

To describe our sample study group further, we completed a leg-length measurement and activity profile. Leg length was

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Fig 1. The DLLT.

measured from the anterior superior iliac spine (ASIS) to the medial malleolus of the same leg.⁹ The mean leg length was 94.3 ± 5.2 cm for the men and 86.2 ± 3.7 cm for the women. The activity profile was a self-reported description of the subjects' activity levels and identified their participation in abdominal strengthening exercises.¹⁰ Eighty-eight percent of both sexes self-reported being either moderately or highly active physically. Eighty-two percent of the men and 76% of the women engaged in a regular program of physical activity. Fifty-two percent of the men and 50% of the women had a regular program of abdominal strengthening.

Double Leg-Lowering Measurement

All DLLT measurements were completed by a team of 2 examiners. Several researchers^{4,7} have described various leg-lowering maneuvers to test abdominal muscle performance. We selected the DLLT as described by Kendall et al,⁷ in which abdominal muscles are used to maintain the pelvis in a posterior-tilted position while the extended legs are lowered from a vertical position. Subjects wore shorts and removed their shoes to avoid additional external loads. The examiners explained the testing procedure to the subjects, who then were allowed to practice the procedure only once to demonstrate their understanding of the DLLT. When performing the test, each subject lay supine on a wooden table with a 1-cm-thick felt pad with the arms folded across the chest. Two trials were performed with a 1-minute rest between trials.

Examiner 1

The test began with an examiner helping the subject to place his/her legs in a vertical position with the knees extended to the terminal range as allowed by the flexibility of the hamstrings. Each subject was instructed to keep the pelvis posteriorly rotated, so the lumbar spine was held firm to the table, while slowly lowering the legs to a horizontal position. Examiner 1 monitored the position of the low back from the subject's right side by placing fingers between the low back (L4-5 area) and the table. We deviated from the Kendall technique by counting aloud to pace the rate of leg lowering. Counting was paced at a rate so it would take approximately 10 seconds to lower the legs from a 90° vertical position to a 0° horizontal position.^{3,8} Examiner 1 verbally signaled examiner 2 when the subject's back began to lift from the monitoring fingers; this represented the end of the test.

Examiner 2

Examiner 2 recorded the subject's performance with a digital inclinometer.^a A 1-cm, 40-cm-long wooden dowel was fastened to the top of the inclinometer for placement along the long axis of the femur. The inclinometer remained parallel to and in light contact with the subject's left femur during leg lowering (fig 1). At the signal to end the test, a stop button was pressed on the inclinometer to record the position of the legs (in degrees). To assess the accuracy of the inclinometer, 23 randomly known angles generated by a computer graphics plotter were compared with values displayed by the inclinometer. The agreement between the displayed values and the computer-generated angles was perfect.

Data Analysis

Descriptive statistics were calculated for all baseline data. An intraclass correlation coefficient ($ICC_{3,1}$) was calculated to estimate intratester reliability for repeated measures of the DLLT. The mean difference in repeated measurements for the DLLT was calculated in combination with the 95% confidence interval (CI) for the difference, as suggested by Bland and Altman.^{11,12} A linear regression was fitted to the data. The model used the mean angle of the 2 trials as the dependent variable with the following as predictor variables: sex, age, leg length, height, weight, physical activity level, regular program of abdominal strengthening, and number of days a week abdominal strengthening was performed. The proportion of variance explained by the variables in the model was determined by using the R^2 statistic. An independent Student t test was used to determine the presence of a sex difference. A significance level of α equal to .05 was used for all inferential statistics. All data analysis was conducted with the SAS, version 7, statistical package.^b

RESULTS

Descriptive statistics of the 100 subjects are provided in table 1.

The $ICC_{3,1}$ estimating intratester reliability for the repeated measures of the DLLT was .98. Reliability of the repeated measures was described further through a graphic representation proposed by Bland and Altman.^{11,12} For each of the 100 subjects, the mean difference between the value of the first and second measures of the DLLT (y axis) was plotted against the mean value of each of the paired measurements (x axis) (fig 2). The 95% CI of agreement between the first and second measurements was -8.2° to 8.2° .

The mean value $\pm$ standard error of the mean (SEM) for abdominal performance for men was $15.4^\circ \pm 2.3^\circ$ from a horizontal reference; for women, it was $37.0^\circ \pm 3.5^\circ$ (table 2). The difference between men and women on DLLT performance was significant ($t_{98} = -5.3$, $P < .001$). The 95% CI of the difference between the means was -29.6° to -13.3° .

The linear regression model found that sex ($t = -3.2$, $P = .002$) and age ($t = -4.6$, $P < .001$) were significant predic-

Table 1: Descriptive Statistics of the Subjects by Sex

Variable	Men (n=50)		Women (n=50)	
	Mean $\pm$ SD	Range	Mean $\pm$ SD	Range
Age (y)	24.8 $\pm$ 2.4	20–29	22.5 $\pm$ 2.5	19–29
Height (cm)	181.6 $\pm$ 6.6	167.6–195.6	166.4 $\pm$ 6.4	152.4–177.8
Weight (kg)	86.1 $\pm$ 16.2	48.5–131.5	62.0 $\pm$ 7.8	47.6–83.9
Leg length (cm)	94.3 $\pm$ 5.2	82–104	86.2 $\pm$ 3.7	78–93

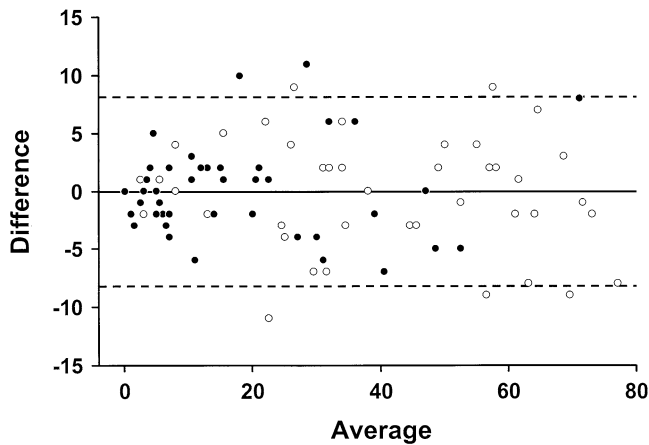


Fig 2. Difference between first and second measurements of the DLLT versus average measurement of DLLT performance for all subjects. Legend: ●, men; ○, women.

tors of DLLT performance (table 3). The R^2 for sex was .22, and for combined sex and age it was .35. Thus, sex accounted for 22% of the variance in DLLT performance and sex and age combined accounted for 35% of the variance. The addition of other variables (leg length, height, body mass, physical activity level, regular program of physical activity, regular program of abdominal strengthening, frequency of abdominal strengthening) did not contribute significantly to the regression. The regression equation is as follows:

$$\hat{y} = 102.7 + 13.3[\text{sex}] - 3.5[\text{age}],$$

where male is 0 and female is 1.

DISCUSSION

We analyzed our reliability based on the following values: .75 and greater, excellent reliability; .40 to .75, fair to good reliability; and less than .40, poor reliability.¹³ Given these values, our calculated value of .98 would be considered excellent reliability. Thus, we showed that intratester reliability with the technique is acceptable for clinical use.

The 95% CI of agreement between the 2 measurements can be used when examining change over time. The 95% interval of agreement represents the range of expected measurement error with the DLLT measurement. Given a measurement error range of $\pm 8.2^\circ$, we would expect that any 2 measures on the same subject would be within $\pm 8.2^\circ$ for 95% of the time or, similarly, that 2 consecutive measurements may be off by up to 8.2° . Clinically, if a person begins a program to enhance abdominal performance, improvement over time on the DLLT greater than 8.2° would represent an improvement in performance.

Zannotti et al¹⁴ questioned the subjectivity of pelvic monitoring because of early rotation of the pelvis in the leg-lowering test. We detected rocking of the pelvis early in leg lowering.

Table 2: Abdominal Muscle Performance of Healthy Men and Women 18 to 29 Years Old Using the DLLT

Subjects	Mean $\pm$ SEM (deg)	Range (deg)
Women	36.9 $\pm$ 3.4	0–77
Men	15.4 $\pm$ 2.3	0–71

Table 3: Linear Regression*

Variable	B	SE	t	P
Constant	102.7	19.30	5.3	.000
Sex	13.3	4.10	3.2	.002
Age	–3.5	0.77	–4.6	.000

*Linear regression model found sex ($t = -3.22$, $P = .002$) and age ($t = -4.57$, $P < .001$) to be significant predictors of DLLT performance.

However, this was followed by a definite holding or stabilization of the pelvis. This phase of the DLLT may be equivalent to the “setting phase” described for the scapula during the initial phases of elevating the upper limb.¹⁵ We believe that this perceived initial rocking can be discerned from when the pelvis begins to rotate anteriorly because of the inability to control external torque demands from the weight of the lower extremities, which indicates the end point of the DLLT. We believe the reliability of our measurement supports the ability to palpate pelvic movement that represents the termination of the test. Others^{16,17} have questioned palpation of the lumbar spine for monitoring performance. Although other landmarks, such as the ASIS could be used, we think that palpation of the lumbar spine provides additional sensory feedback to the subject to reinforce the posterior pelvic positioning and does not alter DLLT performance.

Our multiple regression model indicated that the variables of age and sex were significant predictors of DLLT performance. The negative correlation with age indicates that with increasing age, the measured angle on the DLLT decreases—that is, as age increases, abdominal muscle performance improves. Because of the limited age range (18–29y) of our subjects, we questioned the significance of age as a predictor of performance in our sample. Biering-Sorensen¹⁸ used the DLLT to assess performance in subjects 30 to 60 years old. As with our findings, the difference between men and women in DLLT performance was significant ($P < .001$). However, Biering-Sorensen did not find a significant difference in performance based on age. Youdas et al⁸ also reported a sex difference in mean values for the DLLT, with men on average able to lower their legs to $39.4^\circ \pm 11.3^\circ$ and women to $49.6^\circ \pm 11.5^\circ$. Performance variation within the age range of 30 to 60 years was not reported for either men or women. The values reported by Youdas indicated poorer abdominal performance in comparison with that of our age group, which suggests that at some point performance decreases as age increases.

Previous researchers^{16,17,19} have concluded that the DLLT is a more challenging test for the abdominal muscles than the abdominal curl and have reported distinctly different muscle recruitment patterns in the performance of the 2 tests. In addition, some researchers^{20,21} have proposed that the DLLT offers the advantage of providing a wider potential range of resistance than the abdominal curl, thus potentially allowing for greater discrimination as a test. Minimal external demands are imposed on the abdominal muscles at the start of the DLLT with the legs in a vertical position. However, as the legs are lowered toward a horizontal position, the external torque created by the mass of the limbs reaches its peak, resulting in a challenge to the abdominal muscles that is greater than that imposed with an abdominal curl.

Core strengthening has become popular among fitness and rehabilitation professionals to help a patient stabilize and support the pelvis and spine. Various abdominal muscles have been reported as key muscles in this program, including the external and internal obliques, which also contribute to perfor-

mance on the DLLT.²²⁻²⁴ Other terms for core strengthening are *dynamic stabilization* and *lumbar stabilization*. Although core strengthening is popular, research concerning it is minimal.⁶ Some²⁵ have discussed the use of variations of the DLLT as a more difficult exercise in a progressive lumbar stabilization program. Our results provide evidence for objectively assessing abdominal performance with the DLLT, which may be used to evaluate progress in a core strengthening program. In addition, unlike the action in the trunk curl, abdominal muscles typically function in an isometric stabilizing fashion in the DLLT, making it more representative of a lumbar stabilization program.

Limits of our study include the use of 2 examiners to monitor the position of the pelvis and to record the position of the legs at the instant the lumbar spine lifted from the table. Also, because we studied subjects without back pain, our results may not be generalizable to people with back problems.

Additional studies are indicated to investigate changes over the life span, to discover how performance is related to the incidence of back pain, and to study improvement in performance with specific exercise programs.

CONCLUSIONS

We have found that the DLLT has excellent intratester reliability. Our data provide useful clinical guidelines for assessing abdominal performance in subjects between 18 and 29 years old. We also found a sex difference in abdominal muscle performance as measured by the DLLT, with men able to lower their legs on average to $15.4^{\circ} \pm 2.3^{\circ}$ from a horizontal reference and women able to lower their legs on average to $37.0^{\circ} \pm 3.5^{\circ}$.

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Suppliers

- Digital inclinometer; The Saunders Group, Physical Therapy Equipment, 4250 Norex Dr, Chaska, MN 55318.
- SAS Institute, 100 SAS Campus Dr, Cary, NC 27513.